

be a *minimal* (area) disk mapped to V with $\partial B' = S$. This disk can be represented by a many layer diagram. Namely, assuming the lift of S^1 to the universal covering of V is embedded, all disks in the diagram are mapped into V by ramified coverings over some 2-cells D in V with ramifications of order $\geq k(D)$. Now we give a metric to B' by claiming each k_i -gone in the diagram is isometric to the regular k_i -gone with unit edges in the plane $\mathbb{R}^2$ for the case $K_c \leq 0$ and to such a k_i -gone in the hyperbolic plane of curvature $-\epsilon$ for small $\epsilon > 0$ if $K_c < 0$. This metric has $K \leq 0$ (respectively $K < 0$) for $K_c \leq 0$ ($K_c < 0$) and by a classical (and easy) theorem (see [B-Z])

$$\text{Area } B' \leq (4\pi)^{-1}(\text{length } S)^2 \text{ for } K_c \leq 0,$$

and if $K_c < 0$, then additionally

$$\text{Area } B \leq \text{const length } S.$$

This proof shows that if V has $K_c \leq 0$ and is not hyperbolic, then there exists a partition of $\mathbb{R}^2$ into infinitely many bounded convex k_i -gons and a map $\mathbb{R}^2 \rightarrow V$ satisfying the one layer diagram condition around every k_i -gone of this partition.

It seems to be unknown if the existence of such a map $\mathbb{R}^2 \rightarrow V$ implies for $K_c(V) \leq 0$ that the fundamental group of V contains an isomorphic copy of $\mathbb{Z} \oplus \mathbb{Z}$. There is little evidence in favor of such $\mathbb{Z} \oplus \mathbb{Z}$. In fact, there probably exist compact semihyperbolic (e.g. with $K \leq 0$) spaces receiving $\mathbb{R}^2$ but having no $\mathbb{Z} \oplus \mathbb{Z}$ in π_1 . Some geometers believe that *closed manifolds* with $K \leq 0$ and with non-hyperbolic π_1 do have $\mathbb{Z} \oplus \mathbb{Z} \subset \pi_1$ but well known examples of non-periodic (and non-periodizable) tilings of $\mathbb{R}^2$ by congruent domains make such belief ill founded. There is a similar problem in complex geometry. Let a compact complex algebraic space (or manifold) V receive a non-constant map $\mathbb{C} \rightarrow V$. Does V receive a non-constant map of some compact complex torus?

§5. Basic properties of word hyperbolic groups

In the following sections we denote by Γ a finitely generated group with a fixed δ -hyperbolic word metric.

Let us apply the discussion in 3.1 to the infinite cyclic subgroup $C_\gamma \subset \Gamma$ generated by a non-torsion element $\gamma \in \Gamma$. Then we see that there are two fixed points, called the *poles* γ^∞ and $\gamma^{-\infty}$ in $\partial\Gamma$ defined by

$$\gamma^\infty = \lim_{i \rightarrow \infty} \gamma^i \text{ and } \gamma^{-\infty} = \lim_{i \rightarrow -\infty} \gamma^i.$$

Denote by $M_\gamma \subset \Gamma$ the subgroup of transformations which map the subset $\{\gamma^\infty, \gamma^{-\infty}\} \subset \partial\Gamma$ onto itself. It is clear that M_γ is an elementary group and that M_γ contains the normalizer of $C_\gamma \subset \Gamma$. In fact, M_γ is the (unique) maximal finite extension of C_γ in Γ . It follows, that $\gamma = \gamma_0^p$ for some *exponent* $p = p(\gamma) \in \mathbb{N}$ and some *prime* element $\gamma_0 \in M_\gamma$. If M_γ has no torsion, then γ_0 is unique. Namely γ_0 is the generator of $M_\gamma \approx \mathbb{Z}$ having the same sign (in $\mathbb{Z}$) as γ .

The action of γ on $\partial\Gamma$ is very simple: there are fundamental systems of neighborhoods $U_1^+ \subset \partial\Gamma$ of γ^∞ and $U_1^- \subset \partial\Gamma$ of $\gamma^{-\infty}$, such that γ^i maps $\partial\Gamma \setminus U_1^-$ to U_1^+ for $i = 1, 2, \dots$.

All this is well known in the geometric context and the case of a general word hyperbolic group is equally obvious (see 8.1.).

5.1 Density of poles: Denote by $\partial^2\Gamma \subset \partial\Gamma \times \partial\Gamma$ the set of pairs of *distinct* points in $\partial\Gamma$ and let $P(G) \subset \partial^2\Gamma$ denote the subset of pairs $(\gamma^\infty, \gamma^{-\infty}) \in \partial^2\Gamma$ for all non-torsion elements $\gamma \in G$, where $G \subset \Gamma$ is a given subset.

5.1.A (See 8.2.) *If G contains at most finitely many pairwise non-conjugate elements in Γ then $P(G)$ is discrete. If G is a cobounded subset in Γ (e.g. $G = \Gamma$), then $P(G)$ is dense in $\partial^2\Gamma$.*

5.1.B Corollary: *Every non-elementary word*

hyperbolic group Γ contains infinitely many conjugacy classes of prime non-torsion elements.

5.2 Markov coding in Γ : Consider a finite set G and a set H of finite sequences (words) of elements $g \in G$. Call H *Markov* if there exist a finite set S (of states) with a preferred element $s_0 \in S$, a subset $T \subset S \times S$ (of transitions) and a (labeling) map $\alpha: T \rightarrow G$, such that H equals the set of sequences of the form

$$\alpha(s_0, s_1), \alpha(s_1, s_2), \dots, \alpha(s_{k-1}, s_k)$$

for all sequences $s_0, s_1, \dots, s_k \in S$, such that $(s_i, s_{i+1}) \in T$ for $i = 0, \dots, k-1$ and for all $k = 1, 2, \dots$.

Now, let G be a finite symmetric generating subset in a group Γ ; that is, $G^{-1} = G$. A word w in $g \in G$ representing some $\gamma \in \Gamma$ is called *shortest* if $\text{length } w = |\gamma|$ for the word metric associated to G . Next, fix a linear ordering in G and call a shortest w representing γ *normal* if it is minimal for the lexicographic order on the set of the shortest words representing γ .

5.2.A Theorem: (See 8.5.) *If Γ is word hyperbolic then the sets of the shortest and of the normal words are Markov.*

5.2.A' Corollary: (Compare [Can].) *Let N_k denote the number of $\gamma \in \Gamma$ with $|\gamma| = k$. Then the counting function $Z(t) = \sum_{k=0}^{\infty} N_k t^k$ is rational. Similarly the counting function for the number of shortest words in Γ of length $= k$ is rational.*

5.2.B Fix some $\gamma_0 \in \Gamma$ and let $N_k(\gamma_0)$ denote the number of $\gamma \in \Gamma$ conjugate to γ_0 and having $|\gamma| = k$.

Theorem: (See 8.5.) *The function $\sum_k N_k(\gamma_0) t^k$ is rational for all $\gamma_0 \in \Gamma$.*

This implies with 2.2.B that the functions counting torsion and non-torsion γ with $|\gamma| = k$ separately are rational.

5.2.C Define the *norm* of the *conjugacy class* $[\gamma]$ of $\gamma \in \Gamma$ by

$$|[\gamma]| = \inf_{\gamma' \in [\gamma]} |\gamma'|$$

and let

$$|[\gamma]|_- = \liminf_{n \rightarrow \infty} n^{-1} |[\gamma^n]|.$$

We shall see in 8.5 that there exists an integer m_0 depending on

(Γ, G) , such that $|[\gamma]|_- = m_0^{-1} |[\gamma^{m_0}]|$ for all $\gamma \in \Gamma$. Now, write each non-torsion $\gamma \in \Gamma$ as γ_0^p for some prime γ_0 and $p > 0$, let $\mu[\gamma] = p^{-1} |[\gamma]|_-$ and denote by $[N]_k$ the sum $\sum_{\substack{[\gamma] \\ \text{non-torsion } \gamma \in \Gamma}} \mu[\gamma]$ over the classes $[\gamma]$ with $|[\gamma]|_- = k/m_0$ for all non-torsion $\gamma \in \Gamma$.

5.2.D Theorem: (See 8.5.) *The counting function $\sum_k [N]_k t^k$ is rational.* Furthermore

$$[N]_k \geq (1+\epsilon)^k \text{ for some } \epsilon > 0$$

unless the group Γ is elementary.

5.2.E Remark: A group Γ generated by $\gamma_i \in \Gamma$, $i = 1, \dots, k < \infty$ is called *Markov* if there exists a Markov set H of (not necessarily shortest) words in γ_i , such that each $\gamma \in \Gamma$ is represented by a unique $h = h(\gamma) \in H$. Clearly, "Markov" is invariant under isomorphisms of groups and stable under free and Cartesian products of groups. Moreover, every extension of a Markov group by another Markov group is a Markov group. Apparently, little else is known about Markov groups.

5.3 Free and non-free subgroups in Γ : We have seen in 3.1 how to detect freedom of some subgroups Γ' in Γ by looking at the action of Γ' on $\partial\Gamma$. Now we want exploit another geometric concept, namely the idea of convexity.

Definition: A subset Y in a geodesic space X is called ϵ -convex (or *quasiconvex*) if every geodesic segment in X with the end points in Y lies ϵ -close to (i.e. in the ϵ -neighborhood of) Y . If Γ is a group with a word metric then the notion of quasiconvexity applies to the Cayley graph $X \supset \Gamma$. Namely, $\Gamma' \subset \Gamma$ is quasiconvex in Γ iff it is quasiconvex in X .

If Γ is word hyperbolic and Γ' is a quasiconvex subgroup in Γ , then, by an easy argument, Γ' is finitely generated and word hyperbolic. Furthermore, the embedding $\Gamma' \hookrightarrow \Gamma$ extends to a topological embedding $\partial\Gamma' \rightarrow \partial\Gamma$. Also notice that the quotient space Γ/Γ' with the metric induced from the word metric in Γ (where Γ' acts from the left and hence, isometrically on Γ) is hyperbolic for each quasiconvex subgroups Γ' in every word hyperbolic group Γ .

An important fact is (see Section 7) the *locality* of quasiconvexity.

Geometric example: Let X be a convex manifold with $K \leq -1$ and let $Y \subset X$ be a connected *geodesic graph* in X . That is Y consist of geodesic segment joining some pairs in a discrete set $Y^0 \subset X$ of the vertices in Y . Suppose all (geodesic) edges in Y have length $\geq \ell$ for a fixed $\ell > 0$ and every two edges at each vertex in Y^0 have angle $\geq \alpha > 0$. A simple application of CAT shows that:

If $\alpha \exp \ell \geq 10$, then Y_0 is a quasiconvex tree in X .

A good way to see why some length-angle inequality prohibits any closed curve $Y' \subset Y$ is to take the minimal ball $B \supset Y'$ in X around a fixed point. If Y' is closed, we get at some point $b \in \partial B$ two edges in B starting from b . Then by CAT, there exists a ball B' in the hyperbolic plane and two edges in B' starting from some $b' \in \partial B'$ and having the same lengths and the angle as the original

edges in B . This gives a (sharp for $\text{Rad } B' \rightarrow \infty$) bound from above on the angle in terms of the lengths of the edges.

An algebraic application: Let X be the universal covering of a manifold V_0 with fundamental group Γ_0 and let Γ_0 be generated by a map f of some graph (e.g. a Cayley graph of Γ_0) P into V . That is $f_*: \pi_1(P) \rightarrow \pi_1(X)$ is surjective. Such an f can be made *geodesic* by replacing the map at every edge of P by a locally isometric (i.e. geodesic) map. The lift of this map to X is a (mapped rather than embedded) geodesic graph in X . The edges in Y and the angles between them are the same as for $f(P) \subset X$. Hence, under a suitable edge-angle inequality Y is a quasiconvex tree and consequently Γ_0 is free. Furthermore, if V_0 is a covering of a compact manifold V with $\pi_1(V) = \Gamma \supset \Gamma_0$, then Γ_0 is quasiconvex in Γ .

In order to achieve a suitable edge-angle inequality it is useful to take (P, f) which minimizes some geometric characteristic of P and f . For example, one can let P be homotopy equivalent to the wedge of k circles where k is the number of generators of a given subgroup $\Gamma_0 \subset \Gamma$ and then take map $f: P \rightarrow V$ of *minimal length* generating $\Gamma_0 \subset \Gamma$ (i.e. $f_*(\pi_1(P)) = \gamma \Gamma_0 \gamma^{-1}$ for some $\gamma \in \Gamma$). Then one sees that Γ_0 free and quasiconvex in Γ , provided the injectivity radius of V_0 satisfies

$$\text{Rad } V_0 \geq 100 \log(k+10),$$

where $\text{Rad } V_0$ is defined as the upper bound of those R for which every ball of radius R in X *injects* into V_0 under the covering map $X \rightarrow V$.

In the special case of $k = 2$ one can additionally benefit by looking at two types of graphs $P \rightarrow V_0$

- (1) two closed geodesics joined by an edge.
- (2) a closed geodesic and an edge with the end points on this geodesic.

Then one takes the shortest graph P among all these and observe that the corresponding Γ_0 is free convex unless the total length ℓ of the shortest P is ≤ 10 .

These considerations and the quasiconvex discussion in Section 7 lead to the following freedom criteria for subgroups Γ_0 generated by k elements in a word δ -hyperbolic group Γ .

5.3.A *If $\text{Rad } \Gamma_0$ (defined below) satisfies $\text{Rad } \Gamma_0 \geq \delta 1000 \log(k+100)$ then Γ_0 is free and quasiconvex in Γ .*

5.3.B *If $k = 2$, and Γ has no torsion then Γ_0 is free quasiconvex unless it is generated by $\gamma'_1 = \gamma\gamma_1\gamma^{-1}$ and $\gamma'_2 = \gamma\gamma_2\gamma^{-1}$ for some γ, γ_1 and γ_2 in Γ , such that $|\gamma_1| \leq 100\delta$ and $|\gamma_2| \leq 100\delta$. Thus Γ contains at most finitely many conjugacy classes of non-free subgroups Γ_0 generated by two elements.*

Now let us define $\text{Rad } \Gamma_0$. First recall that $|\gamma| = \inf |\gamma'|$ for all $\gamma' \in \Gamma$ conjugate to γ and then set $\text{Rad } \Gamma_0 = \frac{1}{2} \inf_{\gamma \in \Gamma_0^*} |\gamma|$ for $\Gamma_0^* = \Gamma_0 \setminus \{\text{id}\}$.

Example: Let $\Gamma^1 \supset \Gamma^2 \supset \dots$ be a sequence of subgroups of finite index in Γ . Then $\text{Rad } \Gamma^i \rightarrow \infty$ for $i \rightarrow \infty$ if and only if

$\bigcap_i \Gamma^i = \{\text{id}\}$. Thus 5.3.A applies to subgroups $\Gamma_0 \subset \Gamma^i$ for large i .

Remark: Probably, "generic" word hyperbolic groups admit no such sequences of subgroups of finite index.

5.3.C Let us indicate a simple (partial) generalization of 5.3.B. Start with a quasiconvex (but not necessarily free) subgroup $\Gamma_0 \subset \Gamma$ of infinite index. Notice (compare 8.1, 8.2) that $\partial\Gamma_0 \subset \partial\Gamma$ is nowhere dense in $\partial\Gamma$ and that Γ contains infinitely many Γ_0 -equivalence classes, where γ_1 and γ_2 in Γ are called

Γ_0 -equivalent if $\gamma_1 = \gamma_0 \gamma_2 \gamma'_0$ for some γ_0 and γ'_0 in Γ_0 . In fact one can define $|\gamma| \bmod \Gamma_0 = \inf_{\gamma'} |\gamma'|$ over all γ' which are Γ_0 -equivalent to γ and observe that the function $\gamma \mapsto |\gamma| \bmod \Gamma_0$ is unbounded on Γ . Now, let $\Gamma_1 \subset \Gamma$ be generated by Γ_0 and some $\gamma \in \Gamma$ and notice that Γ_1 only depends on the Γ_0 -equivalence class of γ . A useful choice is a *minimal* $\gamma \in \Gamma$, where $|\gamma| = |\gamma| \bmod \Gamma_0$. If $|\gamma|$ is large then adding corresponding geodesic edge to (some graph generating) Γ_0 does not hurt quasiconvexity (see Section 7) and therefore

5.3.C₁ *For all but finitely many Γ_0 -equivalence classes of γ the subgroup $\Gamma_1 \subset \Gamma$ is quasiconvex and isomorphic to the free product $\Gamma_0 * C_\gamma$ for the cyclic group generated by γ .*

Corollary: Take a random sequence of elements γ_i , $i = 1, 2, \dots$ with large and fast growing length $|\gamma_i|$. Then the subgroups $\Gamma_i \subset \Gamma$ generated by $\gamma_1, \dots, \gamma_i$ are free quasiconvex, unless Γ is elementary. Thus the subgroup $\Gamma_\infty = \bigcup_i \Gamma_i \subset \Gamma$ also is free (but not quasiconvex as it is infinitely generated).

5.3.D Remark: It seems that subgroups Γ_0 as in 5.3.A having $\text{Rad } \Gamma_0$ large compared to the number of generators of Γ_0 have free normal closures $\mathcal{N}(\Gamma_0) \subset \Gamma$. If Γ is non-elementary one also suspects there exists an epimorphism of Γ onto a non-elementary word hyperbolic group such that the kernel is a free subgroup in Γ containing Γ_0 . This is motivated on one hand by the theory of one relator groups and on the other by some results for $\text{SL}_2\mathbb{C}$ in [Lub]. Additional evidence comes from the following

5.3.E Theorem: *There exists a constant $m = m(k, \delta)$, such that for every k hyperbolic elements $\gamma_1, \gamma_2, \dots, \gamma_k$ in a word δ -hyperbolic group Γ the normal subgroup generated by $\gamma_i^{m_i}$, $i = 1, \dots, k$, is free for all $m_i \geq m$.*

The idea of the proof is clear from the following

5.3.E₁ Example: Let V be a complete manifold with $K(V) < 0$ and let $\gamma \subset V$ be a closed geodesic. Then there is no non-trivial relation in $\pi_1(V)$ between the conjugates of the elements represented by γ^{m_i} for $m_i \geq m$, where m is some constant depending on V and γ .

Proof: Every normal relation between various powers γ^{m_i} is given by a map f of a surface F of genus zero into V such that each component of the boundary, say $S_i \subset \partial F$, for $i = 1, \dots, p$, goes onto γ with non-zero degree divisible by m_i . If the relation associated to the map $f: (F, \partial F) \rightarrow (V, \gamma)$ is formally irreducible, then (without any assumption on $K(V)$) the area of F in V (obviously) is bounded from below by $\epsilon \sum_{i=1}^p m_i$, where $\epsilon > 0$ depends on (V, γ) . Now, we take a *minimal* surface F' homotopic to F (as a map $(F, \partial F) \rightarrow (V, \gamma)$) and observe that the induced curvature in F' is $\leq K(V)$. If $K(V) \leq -\epsilon' < 0$ for $\epsilon' > 0$ we have by Gauss-Bonnet.

$$\begin{aligned} p - 1 &= -\chi(F) = -(2\pi)^{-1} \int_F K(F') \geq \epsilon'' \text{Area } F \geq \\ &\geq \epsilon \text{const} \sum_{i=1}^p m_i \geq \epsilon \text{const } pm. \end{aligned}$$

Hence, no such F' exists for $p \geq 2$ and $m > (p-1)(p\epsilon\epsilon')^{-1} \geq (2\epsilon\epsilon')^{-1}$. Q.E.D.

Remarks:

(a) The above argument works for $K \leq 0$ if $K < 0$ near γ . However it completely fails for locally symmetric spaces V with finite volume and rank ≥ 2 (i.e. $K \leq 0$ and nowhere < 0). Indeed by a famous theorem of Margulis every non-trivial normal subgroup in $\pi_1(V)$ has finite index.

(b) There are two possible generalizations of this geometric

argument. First, one can allow surfaces F of positive genus and thus give a lower bound on the genus. Algebraically speaking, surfaces of genus q represent relations between (conjugates of) γ and q commutators in $\Gamma = \pi_1(V)$. The second generalization concerns relation between parabolic elements γ in (non-word) hyperbolic groups, such as fundamental groups of complete non-compact manifolds V with $K < 0$ and with finite volume. Here one uses non-compact minimal surfaces in V properly mapped into V . The area of such a surface can be rather sharply bounded from below by the size of cusps in V which shows in many cases parabolic elements to be normally independent.

(c) Instead of minimal surfaces one can equally use surfaces subdivided into Thurston's hyperbolic geodesic triangles. Since such triangles exist for every word hyperbolic group (see 8.3) our geometric argument proves 5.3.E as well as 5.3.E₁.

(d) Instead of powers of fixed elements γ_i one could use arbitrary sufficiently long closed geodesics (or conjugacy classes in the algebraic context) satisfying some *geometric small cancellation* condition claiming that no geodesic among γ_i follows too closely another geodesic for a considerable amount of time. (See 8.3 for precise statements of this kind.)

5.3.F Non-free subgroups in Γ : If Γ is an arithmetic subgroup in an algebraic group A , say defined over $\mathbb{Q}$, then one finds plenty on non-free subgroups in Γ corresponding to $\mathbb{Q}$ -subgroups in A . Apart from such Γ there is no systematic way to locate non-free subgroups, though many can be seen in the examples in Section 4. Yet one may suspect there exist word hyperbolic groups Γ with (arbitrarily) large $\dim \partial\Gamma$ (here, large is ≥ 1) where every proper subgroup is free.

Here is a general finiteness result for non-free subgroups which we state, for simplicity's sake, for *torsion free* word hyperbolic groups Γ .

5.3.C' Theorem: Fix a non-abelian finitely generated group Γ_0 which is not a non-trivial free product. Then Γ contains at most finitely many conjugacy classes of subgroups isomorphic to Γ_0 .

To see the idea we assume as earlier $\Gamma = \pi_1(V)$ for $K(V) \leq -\epsilon < 0$, let Γ_0 be the fundamental group of a compact (which makes Γ_0 finitely presented) simplicial polyhedron P of dimension two and consider a continuous map $f_0: P \rightarrow \Gamma$ representing a conjugacy class (in Γ) of a homomorphism $\Gamma_0 \rightarrow \Gamma$. Then we use a simple but very powerful

Lemma of Thurston: The map f_0 is homotopic to a map f , such that the area of (the induced metric in) P is bounded by the topology of P ,

$$\text{Area}_f P \leq \pi n_2 \epsilon^{-2},$$

where n_2 is the number of 2-simplices in P .

Proof: "Straighten" every two simplex of P in V by first homotoping the edges to (locally) geodesic segments in V and then fill in the boundaries of 2-simplices by minimal disks. We see as earlier with the Gauss-Bonnet theorem, that the new 2-simplices have area $\leq \pi \epsilon^{-2}$ and the lemma is proven.

Now, with the bound on area P one easily finds a "short" systems of generators in $\Gamma_0 = \pi_1(P)$ unless Γ_0 decomposes into a free product (see 6.7 in [Gr3] for details) and with our geodesic simplices in 8.3 the geometric argument applies to all word hyperbolic groups.

Example: Let Γ be presented by

$$\langle g_1, g_2, g_3 \mid g_3 g_1 g_3^{-1} = w_1, g_3 g_2 g_3^{-1} = w_2 \rangle$$

where w_1 and w_2 are "sufficiently long" words in g_1 and g_2 , such that the 1/6-condition is satisfied and Γ is hyperbolic. Then we have an

infinite sequence of subgroups $\Gamma_i \subset \Gamma$ generated by g_1, g_2 and g_3^1 . It is easy to see that these Γ_i are all word hyperbolic with connected boundaries $\partial\Gamma_i$ of $\dim \partial\Gamma_i = 1$. These groups are not pairwise conjugate as they have different images in $\Gamma/[\Gamma, \Gamma]$ and hence, there are among them infinitely many that are pairwise non-isomorphic. Probably, it is not hard to see directly that all Γ_i are pairwise non-isomorphic (with something like the Alexander polynomial).

Remark: In this example every Γ_i is generated by three elements. No such example is possible with two generators according to 5.3.B. On the other hand cyclic coverings of the ramified torus fibered over S^1 (see 4.4) give us geometric examples with rather large number of generators.

5.4 Monomorphisms into hyperbolic groups: Thurston's lemma in 5.4.C was generalized and refined by Thurston for maps of polyhedra of dimension ≥ 2 into spaces with $K < 0$. This allows one to sharpen in some cases the finiteness theorem in 5.3.E by replacing conjugation in Γ by conjugation in Γ_0 . Here is an instance of such a result which is due to Thurston in the geometric case (compare [Th], [Gr4]).

5.4.A *Let V be a closed aspherical (i.e. the universal covering is contractible) manifold with a word hyperbolic fundamental group Γ . If $\dim V \geq 3$ then the exterior automorphism group $\text{Aut } \Gamma / \text{Int } \text{Aut } \Gamma$ is finite.*

To grasp the idea let us show that the image of $\text{Aut } \Gamma$ in $\text{Aut}(\Gamma/[\Gamma, \Gamma])$ is finite. Now, $\Gamma/[\Gamma, \Gamma] = H_1(V)$ which is the Poincaré dual to $H^{n-1}(V)$ for $n = \dim V$. Recall that the cohomology $H^*(V; \mathbb{R})$ carries a natural pseudo-norm (which may be infinite) corresponding to the L_∞ -norm on the space of singular cochains (see [Gr2]). An application of Thurston's lemma shows this pseudo-norm is an actual (i.e. $< \infty$) norm on $H^{k \geq 2}(V)$ if $K(V) < 0$ and a similar fact for $H^{k \geq 2}(\Gamma; \mathbb{R})$ holds true for all word hyperbolic groups Γ by the properties of the geodesic flow on $\hat{G}(\Gamma)$ (see 8.3). The norm on

$H^*(\Gamma)$ (obviously) is unvariant under the action of $\text{Aut } \Gamma$, which implies the finiteness of the image of $\text{Aut } \Gamma$ in $H^{k \geq 2}(\Gamma; \mathbb{R})$. If $n = \dim V \geq 3$ then, by the Poincare duality, the image of $\text{Aut } \Gamma$ in $H_1(\Gamma)$ also is finite.

5.4.B Another standard application of Thurston's simplices (which are constructed for word hyperbolic groups in 8.3) is the following

Theorem: *Let V be a compact orientable aspherical pseudomanifold with word hyperbolic fundamental group Γ and let V' be an arbitrary compact orientable pseudomanifold with $\dim V' = n = \dim V$. Then there exists a constant $d = d(V, V')$, such that the degree of every continuous map $f: V' \rightarrow V$ is bounded by $|\deg f| \leq d$. (Compare [Gr2] [Gr4].)*

By applying this to iterates of a selfmapping $V \rightarrow V$ one concludes that Γ is not isomorphic to any of its proper subgroups. Probably, the same is true for every word hyperbolic group Γ connected at infinity.

5.4.C It seems very plausible that Thurston's theory of Kleinian groups can be set up in a general hyperbolic context. Namely, we start with an arbitrary finitely generated group Γ and try to describe the Teichmüller space T (or at least some ideal boundary ∂T) of all (non-word) δ -hyperbolic left invariant metrics on Γ whose "curvature" is bounded from below as well as from above (by $-\delta^{-2}$). For example, one may look at complete simply connected manifolds of a fixed dimension with $-\delta^{-2} \geq K(X) \geq -C$ which are discretely and isometrically acted upon by Γ . The pertinent metrics on Γ are induced by orbit maps $\gamma \mapsto \gamma x$ for some $x \in X$ and metrics associated to different $x \in X$ should correspond (with an appropriate definition) to same point in the Teichmüller space $T(\Gamma)$. In particular, we do not distinguish metrics on Γ which are equivalent under inner automorphisms of Γ . Yet we must keep track of outer automorphisms. Otherwise we get Riemann moduli space

rather than the Teichmüller space. Thurston's theory applies to the case $\dim X = 3$ and $K(X) = -1$.

One of the basic results (see [Th] and [M-S]) is the compactness of the space T under certain conditions on Γ which rule out, for example, surface groups Γ . Notice that Thurston's compactness theorem (which generalizes Mostow rigidity theorem) implies the finiteness of $\text{Aut } \Gamma / \text{Int Aut } \Gamma$ as well as the compactness of the Riemann moduli space. Notice that theorem 5.3.C' also can be regarded as a (rather weak) compactness theorem for the Riemann moduli space of Γ_0 .

The first step in the study of $T(\Gamma)$ was made by J. Morgan and the author who found (unpublished) some higher dimensional versions of Thurston compactness theorem. It seems that non-compactness is related to low dimensional (zero and one) pattern in $\partial\Gamma$, in the case Γ is word hyperbolic. (The "zero-dimensional reason" for non-compactness is disconnectedness of $\partial\Gamma$ and the 1-dimensional source of non-compactness probably is due to a presence zero dimensional subsets A in open connected subsets $U \subset \partial\Gamma$, such that $U \setminus A$ is disconnected.) Yet nothing is known about $T(\Gamma)$ in the non-compact case beyond Thurston-Teichmüller theory. One does not even know the structure of the moduli space and (or) of the exterior automorphism groups of fundamental groups of 2-polyhedra with $K < 0$.

5.5 Factor groups of hyperbolic groups: Let Γ be a non-elementary word hyperbolic group and $F \subset \Gamma$ a finite subset. We shall see in this section that there are many hyperbolic factor groups Γ' of Γ for which the quotient map $\Gamma \rightarrow \Gamma'$ is injective on F .

5.5.A Theorem: *Let $\Gamma_i \subset \Gamma$, $i = 1, \dots, k$, be non-elementary subgroups in Γ . Then there exists a factor group Γ' with the following four properties.*

- (1) Γ' is a non-elementary word hyperbolic group.
- (2) The canonical epimorphism $\Gamma \rightarrow \Gamma'$ is injective

on the subset $F \subset \Gamma$.

(3) The epimorphism $\Gamma \rightarrow \Gamma'$ sends Γ_i onto Γ' for all $i = 1, \dots, k$.

(4) The epimorphism $\Gamma \rightarrow \Gamma'$ is bijective on the sets of conjugacy classes of non-free (i.e. not infinite cyclic) elementary subgroups in Γ and Γ' . In particular, if Γ is torsion free then also Γ' is torsion free.

An immediate corollary is the following generalization of a theorem by Olshanski (see [Ol]).

5.5.B Let Γ be a torsion free non-elementary word hyperbolic group (e.g. Γ is free with two generators). Then there exists a non-Abelian quotient group $\bar{\Gamma}$ of Γ , such that every proper subgroup in $\bar{\Gamma}$ (proper means $\neq \bar{\Gamma}$ or $\{id\}$) is infinite cyclic.

5.5.C Consider an infinite elementary subgroup $E_0 \subset \Gamma$ and let $C \subset E_0$ be a maximal infinite cyclic normal subgroup. Then the subgroup mC is normal in E_0 and we define $E_m = E_0/mC$ for all $m = 1, 2, \dots$.

5.5.D Theorem: There exists an integer $m = m(\Gamma, E_0, F)$ such that for every $n \geq m$ there exists a factor group Γ' of Γ with the following four properties.

- (1) Γ' is non-elementary word hyperbolic.
- (2) The homomorphism $\Gamma \rightarrow \Gamma'$ is injective on F .
- (3) The image E'_0 of E_0 in Γ' is isomorphic to E_n .

(4) Every torsion element γ' in Γ' is conjugate to some $\gamma'' \in E_0'$ or it is the image of a torsion element $\gamma \in \Gamma$.

By combining 5.5.A and 5.5.D we immediately obtain the following

5.5.E Corollary (Compare [Ol]): Let Γ be a torsion free non-elementary word hyperbolic group. Then there exists a sequence of integers $m_1, m_2, \dots$, such that for every sequence $n_i \geq m_i$, $i = 1, 2, \dots$ there exists an infinite factor group $\bar{\Gamma}$ of Γ containing subgroups $\bar{\Gamma}_i \subset \bar{\Gamma}$, $i = 1, 2, \dots$ with the following three properties.

- (1) $\bar{\Gamma}_i \approx \mathbb{Z}/n_i\mathbb{Z}$ for $i = 1, 2, \dots$.
- (2) $\bar{\gamma}\bar{\Gamma}_i\bar{\gamma}^{-1} \cap \bar{\Gamma}_j = \{e\}$ for all $\bar{\gamma} \in \bar{\Gamma}$ and all $i \neq j$.
- (3) Every proper subgroup $\bar{\Gamma}'$ in $\bar{\Gamma}$ is conjugate (in $\bar{\Gamma}$) to a subgroup in $\bar{\Gamma}_i$ for some $i = i(\bar{\Gamma}')$. In particular, $\bar{\Gamma}$ is a torsion group.

Remark: We shall see later that the sequence $m_i = m_i(\Gamma)$ can be chosen primitively recursive, but our proof does not allow bounded sequences m_i . One believes however, that every non-elementary word hyperbolic group Γ admits an infinite quotient groups whose elements have finite orders bounded by a constant $m = m(\Gamma)$.

5.5.F About the proofs: Everything boils down to showing that adding "sufficiently random" relations to a non-elementary word hyperbolic group gives us a word hyperbolic group again, and short words in the original group are not affected by the relations. To see this consider some closed geodesics in a manifold V and let

$L_1, L_2, \dots$ be the connected components of the lifts of these geodesics to the universal covering X of V . We assume as earlier that X is convex with $K \leq -1$ and then observe that each L_i is an isometrically embedded (geodesic) line in X . (Notice that every geodesic in V usually lifts to infinitely many lines in X .) Fix a number $\epsilon > 0$ and call a segment $[a, b] \subset L_i$ a *piece* if there exists a segment $[a', b']$ in some L_j for $j \neq i$ such that $|a - a'| \leq \epsilon$ as well as $|b - b'| \leq \epsilon$. Now we can formulate the usual small cancellation conditions,

$SC_1(\epsilon, 1/k)$: None among our closed geodesics in V can be covered by (projections from X) of k pieces.

Less natural but better suited for the moment is

$SC_2(\epsilon, 1/k)$: The length $|a - b|$ of every piece is less than $1/k$ (length of the underlying closed geodesic).

In what follows, let $\epsilon = 1$ and assume for simplicity's sake that the injectivity radius of V is bounded from below by $\text{Rad } V \geq 1$. Let Γ be the fundamental group of V and let $\bar{\Gamma}$ be the fundamental group of the space $\bar{V}$ obtained from V by attaching disks to given closed geodesics γ_i in V .

Claim: *If the closed geodesics γ_i in V satisfies $SC_2(1, 0.1)$ then $\bar{\Gamma}$ is word hyperbolic.*

Proof: We are going to prove that every contractible closed curve f in $\bar{V}$ bounds a disk D with area $\leq \text{const length } f$. First, we may assume that f lies in $V \subset \bar{V}$. Next we can (obviously) homotope f to a closed geodesic in V such that the area of this homotopy (that is a map $S^1 \times [0, 1] \rightarrow V$) is bounded by $\text{const length } f$. (Notice, that existence of such homotopy solves the conjugacy problem in Γ if V is compact.) We can assume moreover, that the lift to X of this geodesic contains no pieces $[a, b]$ which are ϵ -close to $[a', b'] \subset L_i$

for some lift L_i of one of the given closed geodesic, such that $|a'-b'| \geq 2/3$ (length of the underlying closed geodesic). Otherwise, one could shorten our geodesic in $\bar{V}$ by roughly $|a'-b'|$ and only adding a fixed small amount of area to the spanning disk. Now, with all these assumptions on f we look at a spanning disk $D^2 \rightarrow \bar{V}$ with $\partial D^2 = f$. The part of this disk inside $V \subset \bar{V}$ is a surface F of genus zero bounded by f and multiples $m_j \gamma_j$ of the given geodesics γ_j . If we chose F *minimal* (compare 5.3.E1) we obtain a surface with $K \leq -1$ bounded by geodesics, such that no component $m_j \gamma_j$ of ∂F lies in a small (of order $\epsilon = 1$) neighborhood of a union of n other components for $n \leq 200m_j/3$. It follows that (a small neighborhood of) each $m_j \gamma_j$ contributes at least -10 (where 10 depends on 100) to the total curvature of F . Then, by Gauss-Bonnet, the number N of $m_j \gamma_j$ -component satisfies

$$N = -\chi(F) = -(2\pi)^{-1} \int_F K(F) \geq 10N/2\pi,$$

which makes $N = 0$ and shows that our map (curve) $f: S^1 \rightarrow V$ is constant. Q.E.D.

This argument extends to word hyperbolic groups if "closed geodesics" are replaced by "closed orbit of the geodesic flow" (see 8.3) and instead of minimal F , we use a subdivision of F into ideal triangles bounded by infinite (as they are lifted to X) geodesics which are orbits of the geodesic flow. Now the proof of 5.5.A-D becomes straightforward and details are left to the reader.

Remark: The logical scheme of the results in this section is essentially due to Olshanski (see [Ol]). The simplification we achieve comes, for the most part, from *explicit* use of the notion of hyperbolicity. However, in deeper question (such as construction of various infinite periodic groups with *bounded* torsion exponents) combinatorial techniques by Olshanski can not be matched so far by simple geometrization of ours. Yet, further development of geometric language will, probably, take care of these questions.

5.6 Kazdan's T-groups: Call Γ a T-group if for every generating subset $G \subset \Gamma$ there exists an $\epsilon > 0$ such that every isometric action of Γ on the unit sphere S in the Hilbert space satisfies the following ϵ -fixed point property:

if $|gs_0 - s_0| \leq \epsilon$ for some $s_0 \in S$ and all $g \in G$ then there exists a fixed point of the action of Γ on S .

The existence of infinite T-groups is a non-trivial fact discovered by Kazdan (see [Ka]) who has shown that every lattice in a simple Lie group of $\mathbb{R}$ -rank ≥ 2 satisfies property T. Notice, that these lattices are not word hyperbolic and that the (hyperbolic) lattices in $O(n,1)$ and in $U(n,1)$ never have property T. However, the lattices in $Sp(n,1)$ do satisfy T for all $n \geq 2$ by a theorem of Kostant (see [Ko]). Thus for every $d = 4n - 1$, $n = 2, 3, \dots$, there exists a word hyperbolic T-group Γ with $\partial\Gamma \approx S^d$.

Observe the following two properties of T-groups (see [Ka]).

5.6.A *If Γ satisfies T then every factor group $\bar{\Gamma}$ of Γ satisfies T.*

5.6.B *If an amenable (e.g. abelian) group Γ satisfies T then Γ is finite. In particular, every infinite factor group of a T-group is non-amenable.*

Now we apply 5.5.A to the free product $\Gamma * \Gamma_1$ where Γ_1 satisfies T and obtain the following.

5.6.C Theorem: *Every non-elementary word hyperbolic group Γ admits a quotient group $\bar{\Gamma}$ which is non-elementary word hyperbolic and satisfies the T-property. Furthermore if Γ is torsion free then $\bar{\Gamma}$ can be also assumed torsion free.*

5.6.D Corollary: *The factor group $\bar{\Gamma}$ in 5.5 can*

be made T and hence, non-amenable. (Compare [Ol].)

Remark: We shall see in 5.5 that the group Γ' has

$$\dim \partial\Gamma' = \max(1, \dim \partial\Gamma).$$

But if we want the T-property we only can obtain Γ' with

$$\dim \partial\Gamma' = \max(7, \dim \partial\Gamma)$$

as all known word hyperbolic T-groups Γ have $\dim \partial\Gamma \geq 7$. Probably, there exist word hyperbolic T-groups Γ with $\dim \partial\Gamma = 1$ (or at least, with $\dim \partial\Gamma = 2$). Here is another.

Question: Does there exist for a given integer $d \geq 2$ an infinite word hyperbolic group Γ (e.g. a lattice in $\mathrm{Sp}(d,1)$) whose every infinite word hyperbolic factor group Γ' has $\dim \partial\Gamma' \geq d$? (For $d = 0$ and $d = 1$ the existence of such a Γ is easy.)

5.6.E Non-hyperbolic quotient group: According to the above many word hyperbolic groups Γ (e.g. lattices in $\mathrm{Sp}(2,1)$) have no free factor groups. However, this does not prevent factor groups of Γ from the presence of complicated subgroups as the following theorem shows.

Theorem: Let Γ be a non-elementary word hyperbolic group and Δ an arbitrary finitely generated group. Then there exists a quotient group $\bar{\Gamma}$ of Γ , such that Δ embeds into $\bar{\Gamma}$ and the factor group $\bar{\Gamma}/\mathcal{N}(\Delta)$ is non-elementary word hyperbolic for the normal closure $\mathcal{N}(\Delta)$ of $\Delta \subset \bar{\Gamma}$.

The proof is obtained by a simple modification of the arguments in 5.5 and is left to the reader.

§6. Trees, triangles and surfaces in hyperbolic spaces

In the following sections 6.1-6.6 we relate the δ -inequality to the geometry of geodesic triangles. Then (see 6.7 and 6.8) we prove that the δ -inequality is equivalent to the isoperimetric inequality Is_2 , provided the space in question is geodesic.

Let (X, x_0) be a metric space δ -hyperbolic with respect to x_0 . Consider a sequence of points $x_1, x_2, \dots, x_n$ in X for $n = 2^k + 1$ and observe by induction on k that

$$(*) \quad (x_1, x_n) \geq \min_{1 \leq i \leq n-1} (x_i, x_{i+1}) - k\delta.$$

Notice that every sequence $x_1, x_2, \dots, x_1, \dots, x_n$ can be obviously reduced to a sequence $x'_1 = x_1, x'_2, \dots, x'_j, \dots, x'_m = x_n$, such that $m \leq n$, all x'_j are distinct and $\min_j (x'_j, x'_{j+1}) \geq \min_i (x_i, x_{i+1})$. Thus we conclude to the following

Lemma: *If X is a finite space containing at most $2^k + 2$ points then every finite sequence of points in X satisfies (*).*

6.1 Approximating tree TrX: Let X be δ -hyperbolic for $\delta = 0$ and let us construct a tree $\text{Tr}X$ as follows. Take the disjoint union of segments $[0, |x|] \subset \mathbb{R}$ for all $x \in X$ and then call two points $t_1 \in [0, |x_1|]$ and $t_2 \in [0, |x_2|]$ equivalent if $|t_1| = |t_2|$ and $(x_1, x_2) \geq t_1 = t_2$. Since X is 0-hyperbolic with respect to x_0 , this is indeed an equivalence relation and the quotient space is our tree $\text{Tr}X$. We equip $\text{Tr}X$ with the maximal metric (see 1.3) for which the obvious map $[0, |x|] \rightarrow \text{Tr}X$ is isometric for all $x \in X$ and observe that the map $X \rightarrow \text{Tr}X$ also is isometric. If X is a finite space, then $\text{Tr}X$ is a finite simplicial metric tree and $X \subset \text{Tr}X$ consists of the extremal points in $\text{Tr}X$, where a point x in a tree is called extremal if there is no topological embedding of $\mathbb{R}$ into the tree with $\text{Image } \mathbb{R} \ni x$.

6.1.A Let us assume that every finite sequence of

points $x_1, \dots, x_n$ in X satisfies

$$(x_1, x_n) \geq \min_i (x_i, x_{i+1}) - \delta'$$

for some fixed $\delta' \geq 0$. Then define a new "scalar product" $(x, y)'$ in X as follows. Consider all sequences $\mathfrak{A}$ of points $x_1 = x, x_2, \dots, x_n = y$ and set

$$(x, y)' = \sup_{\mathfrak{A}} \min_i (x_i, x_{i+1}).$$

The corresponding new metric in X ,

$$|x - y|' = |x| + |y| - 2(x, y)',$$

obviously is 0-hyperbolic and satisfies

$$|x| = |x|' \text{ for all } x \in X$$

and

$$|x - y| - 2\delta' \leq |x - y|' \leq |x - y|$$

for all x and y in X .

Thus we conclude to the following

6.1.B Proposition: *Let X be δ -hyperbolic with respect to a reference point $x_0 \in X$ and $\#X \leq 2^k + 2$. Then there exists a map of X into some simplicial metric tree, say $f: X \rightarrow \text{Tr}X$ with the following three properties.*

- (1) $|f(x)| = |x|$ for all $x \in X$.
- (2) The map f is short (i.e. distance non-increasing).
- (3) The codiameter of the map f defined by

$$\text{Codiam } f = \sup_{x, y \in X} (|x-y| - |f(x)-f(y)|)$$

satisfies

$$\text{Codiam } f \leq 2k\delta.$$

6.1.C Remark: As it is clear from the proof the assumption $\#X \leq 2^k + 2$ can be replaced by the following condition:

X admits a decomposition into the union of $2^{k-1} + 1$ subsets $X_i \subset X$, such that each X_i admits an isometric embedding into $\mathbb{R}$ and $x_0 \in X_i$ for all $i = 1, \dots, 2^{k-1} + 1$. (In fact, the condition $x_0 \in X_i$ is not truly essential.)

Then there exists a map of X into a tree, $f: X \rightarrow \text{Tr}X$, which satisfies the above (1), (2) and (3).

6.2 Geodesic trees: Let $Y = \{y_1, \dots, y_n\}$ be a finite subset is a *geodesic* δ -hyperbolic space X and let us sketch a construction of a *geodesic tree* $T \subset X$ with the following three properties.

- (1) T is a union of at most $n-1$ geodesics segments in X .
- (2) The set of extremal points of T equals Y .
- (3) Every two points y_i and y_j in Y can be joined by a broken geodesic g in T containing $k \leq 1 + 2 \log_2 n$ segments and satisfying

$$\text{length } g \leq |y_i - y_j| + C\delta(\log_2 n)^2,$$

for some constant $C \leq 100$. (See Figure 2 for such graphs for $n = 3, 4, 5, 6$)

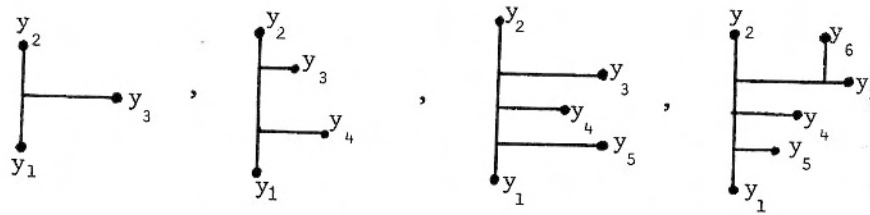


Figure 2

Step 1: Decompose Y into subsets $Y = Y_1 \cup Y_2 \cup \dots \cup Y_m$ for $m \leq \log_2 n$ where Y_1 contains two points and such that for every $i_0 = 1, \dots, m-1$ and for every two points x and y in Y_{i_0+1} there exist

two points x' and y' on the union $U_{i_0} = \bigcup_{i=1}^{i_0} Y_i$, satisfying

$$(x', y')_x + (x', y')_y \leq |x - y| + 4(\log_2 n + 1)\delta$$

where $(x', y')_z$ stands for $1/2(|x' - z| + |y' - z| - |x' - y'|)$. The existence of such Y_i is obvious for trees (i.e. for $\delta=0$) and the general case follows via 6.1.B.

Step 2: Let T_1 be a segment between the two points in Y_1 and define (by induction) a geodesic tree $T_{i+1} \subset X$ with the extremal set $Y_1 \cup Y_2 \cup \dots \cup Y_{i+1}$ as the union of T_i and distance minimizing segments from each point in Y_{i+1} to T_i . One can choose these segments, such that the intersection of any two of them is connected set (usually a single point) containing the T_i -ends of the segments. Then T_{i+1} is indeed a tree and a simple induction on i and on n proves our assertion.

6.3 Thin geodesic triangles: Consider three points x_0, x_1 and x_2 in a metric space X and let T be a (unique up to isometry) simplicial metric tree with three extremal points called x'_0, x'_1, x'_2 such that $|x'_i - x'_j| = |x_i - x_j|$ for $i, j = 0, 1, 2$. (See Figure 3.)

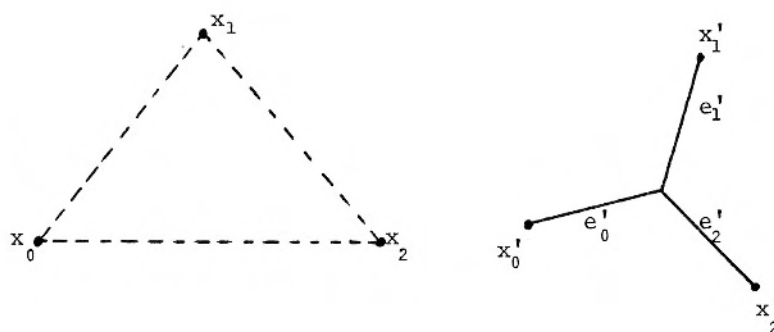


Figure 3

Notice that the edge e'_i in T adjacent to x'_i has length $e_i = (x_j, x_k)_{x'_i}$.

Let Δ be a geodesic triangle with vertices x_0, x_1, x_2 . That is Δ is the union of tree geodesic segments between the pairs of vertices. (Such a Δ always exists, for example, if the space X is geodesic.) The map $\{x_0, x_1, x_2\} \rightarrow \{x'_0, x'_1, x'_2\}$ obviously extends to a unique map $f_\Delta: \Delta \rightarrow T$ which is isometric on each segment of Δ . Say that Δ is δ -thin if $f_\Delta(x) = f_\Delta(y)$ implies $|x - y| \leq \delta$ for all x and y in Δ .

6.3.A Lemma: *If X is δ -hyperbolic then Δ is 2δ -thin.*

Proof: We may assume that x lies on the segment $[x_0, x_1]$ and $y \in [x_0, x_2]$. Consider the map $f: \{x_0, x_1, x_2, x, y\} \rightarrow T$ provided by 6.1.B and observe that $f_\Delta(x) = f_\Delta(y)$ implies $f(x) = f(y)$. Hence 6.1.B implies the lemma.

6.3.B Let us prove a converse to 6.3.A. Consider four point x_0, x_1, x_2 and x_3 in X and geodesic segments $[x_i, x_j]$ between these points.

Lemma: *If the geodesic triangles $[x_0, x_1, x_2]$, $[x_0, x_2, x_3]$ and $[x_0, x_1, x_3]$ are δ -thin then*

$$(x_1, x_2)_{x_0} \leq \min((x_1, x_3)_{x_0}, (x_2, x_3)_{x_0}) - 2\delta.$$

Proof: Let $t_0 = \min((x_1, x_3), (x_2, x_3))$ and let $x'_i \in [x_0, x_i]$, $i = 1, 2, 3$, be some points with $|x_0 - x'_1| = t \leq t_0$. Since the triangles $[x_0, x_1, x_2]$ and $[x_0, x_2, x_3]$ are δ -thin, we have

$$|x'_1 - x'_3| \leq \delta \text{ and } |x'_2 - x'_3| \leq \delta,$$

and then

$$(*) \quad |x'_1 - x'_2| \leq 2\delta.$$

Since $[x_0, x_1, x_2]$ is δ -thin,

$$(**) \quad |x'_1 - x'_2| \geq 2(t - (x_1, x_2)_{x_0} - \delta),$$

for all $t = |x_0 - x'_1| = |x_0 - x'_2|$. Now the lemma follows from (*) and (**) with $t = t_0$.

Let us summarize 6.3.A and B in the following

6.3.C Proposition: *Let X be a geodesic metric space. If X is δ -hyperbolic then all geodesic triangles in X are 2δ -thin. Conversely, if all geodesic triangles in X are δ -thin then X is 2δ -hyperbolic.*

6.4 Hyperbolic geodesic hulls: Start with a three point metric space $X = \{x_1, x_2, x_3\}$ join every pair of points (x_i, x_j) in X by an edge $[x_i, x_j]$ isometric to $[0, |x_i - x_j|] \subset \mathbb{R}$ and equip the resulting triangle Δ with the maximal metric $|\cdot|_{2\delta}$ which agrees with the $[0, |x_i - x_j|]$ -metric on every edge and for which the triangle Δ is 2δ -thin.

Next, for an arbitrary metric space X we take

$$P^1(X) = \bigcup_{d \geq 0} P_d^1(X),$$

where $P_d^1(X)$ is the 1-skeleton of the polyhedron $P_d(X)$ defined in 1.7 and we consider the maximal metric $| \cdot |_{2\delta}$ in $P_d^1(X)$ for which the canonical map $(\Delta, | \cdot |_{2\delta}) \rightarrow (P_1^d(X), | \cdot |_{2\delta})$ is short (i.e. distance non-increasing) for the above $\Delta = \Delta(x_1, x_2, x_3)$ and for all triples of points x_1, x_2 and x_3 in X .

6.4.A Lemma (Compare [Dr]): *If X is δ -hyperbolic then the above map $\Delta \rightarrow P_1^d(X)$ is isometric for all x_1, x_2 and x_3 in X .*

Proof: It is enough to consider the case of a four point space X where the proof is obvious.

6.4.B Corollary: *If X δ -hyperbolic then the canonical embedding $X \rightarrow (P^1(X), | \cdot |_{2\delta})$ is isometric and the space $(P^1(X), | \cdot |_{2\delta})$ is 4δ -hyperbolic.*

6.4.C Let $\bar{\Delta}$ be the 2-simplex bounded by Δ and let us take a specific simplicial metric in $\bar{\Delta}$ for which Δ is thin. Namely, start with the product $\Delta \times [0, 2\delta]$ and then collapse $\Delta \times 2\delta$ to the tripod T as in Figure 3. The resulting space is homeomorphic to $\bar{\Delta}$ and we take the maximal metric as Δ for which the (quotient) map $\Delta \times [0, 2\delta] \rightarrow \bar{\Delta}$ is short. Clearly, every edge of $\Delta = \Delta \times 0 = \partial \bar{\Delta} \subset \bar{\Delta}$ is geodesic for this metric and Δ is 4δ -thin. It is also clear that the induced metric on $\Delta \subset \bar{\Delta}$ is $\geq | \cdot |_{2\delta}$.

Now, let

$$P^2(X) = \bigcup_{d \geq 0} P_d^2(X)$$

and let $| \cdot |_{2\delta}$ be the maximal metric on $P^2(X)$ for which the (canonical) embedding $\bar{\Delta} \rightarrow P^2(X)$ is short for all $\bar{\Delta} = \bar{\Delta}(x_1, x_2, x_3)$. One sees with the above discussion:

6.4.D Proposition: *Let X be δ -hyperbolic.*

Then $(P^2(X), d_{P^2(X)})$ is a δ' -hyperbolic metric space for $\delta' = 12\delta$ and the canonical embedding $X \rightarrow P^2(X)$ is isometric. Furthermore the space $P^2(X)$ is geodesic. In fact, the metric in $P^2(X)$ is simplicial for some subdivision of $P^2(X)$.

6.4.E Remark: The map between the hyperbolic boundaries $\partial X \rightarrow \partial P^2(X)$ induced by the embedding $X \rightarrow P^2(X)$ is a homeomorphism as it easily follows from the construction of $P^2(X)$.

6.5 The inscribed triangle Δ^{in} : Consider a geodesic triangle Δ in X with edges e_i , $i = 1, 2, 3$ and recall the map f_Δ of Δ onto the tree T which is a tripod whose three extremal points correspond to the vertices of Δ (see Figure 3). Then there is a unique triple of points, called $\Delta^{\text{in}} = \{y_1, y_2, y_3\}$ for $y_i \in e_i$, such that all three points go under f_Δ to the central vertex of the tripod T (see Figure 3 and Figure 4 below).

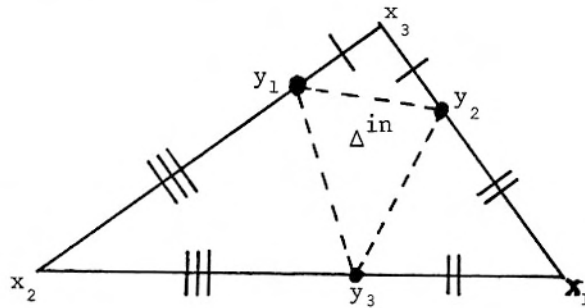


Figure 4

Define

$$\text{insize } \Delta = \text{Diam } \Delta^{\text{in}}$$

and observe that every δ -thin triangle obviously has $\text{insize} \leq \delta$.

6.5.A Suppose that every two points in Δ , say

$y_1' \in e_1$ and $y_2' \in e_2$, satisfying $f_\Delta(y_1') = f'_\Delta(y_2')$ are joined by a geodesic segment $[y_1', y_2']$ in Δ . Denote by Δ' the resulting "subtriangle" $[y_1', y_2', x_3]$.

Lemma: If Δ and all "subtriangles" Δ' have insize $\leq \delta$ then the triangle Δ is 2δ -thin.

Proof: We must show for all $y_1^t \in e_1$ and $y_2^t \in e_2$ with

$$|x_3 - y_1^t| = |x_3 - y_2^t| = t \leq t_0 = (x_1, x_2)_{x_3}$$

that

$$(*) \quad |y_1^t - y_2^t| \leq 2\delta.$$

If $t = t_0$, then

$$(**) \quad |y_1^t - y_2^t| \leq \delta,$$

because insize $\Delta \leq \delta$. Then we define $t_i = t_{i-1} - \delta_{i-1}$ where

$\delta_{i-1} = \min[y_1^{t_{i-1}}, y_2^{t_{i-1}}, x_3]$ for $i = 1, 2, \dots$, and obtain by induction on i the inequality $(**)$ for $t = t_0, t_{i-1}, \dots$. Now $(*)$ follows for all t by the triangle inequality.

6.5.B Corollary: If all geodesic triangles in a geodesic space X have insize $\leq \delta$ then X is 4δ -hyperbolic.

6.6 The minimal size of Δ : Take an arbitrary set Δ in X covered by tree subset $e_i \subset \Delta$, $i = 1, 2, 3$, and consider triples of points $\Delta' = \{y_1, y_2, y_3\}$ in Δ with $x_i \in e_i$ for $i = 1, 2, 3$. Define

$$\text{minsize } \Delta = \inf_{\Delta'} \text{Diam } \Delta'.$$

Observe that minsize does not increase under short maps $\Delta \rightarrow \Delta'$

sending e_i to e'_i , that is

$$\text{minsize } \Delta' \leq \text{minsize } \Delta.$$

Also notice that every geodesic triangle Δ covered by the three edges has

$$\text{minsize } \Delta \leq \text{insize } \Delta.$$

6.6.A Lemma: *Every geodesic triangle Δ satisfies*

$$\text{insize } \Delta \leq 2 \text{ minsize } \Delta.$$

Proof: The position of tree points $y_i \in e_i$ are determined by the six distances d_i^j , for $i \neq j$, to the vertices x_j of Δ satisfying three linear equations

$$(*) \quad \begin{cases} d_1^2 + d_2^3 = |x_2 - x_3| \\ d_2^1 + d_3^3 = |x_1 - x_3| \\ d_3^1 + d_1^2 = |x_1 - x_2|, \end{cases}$$

see Figure 5.

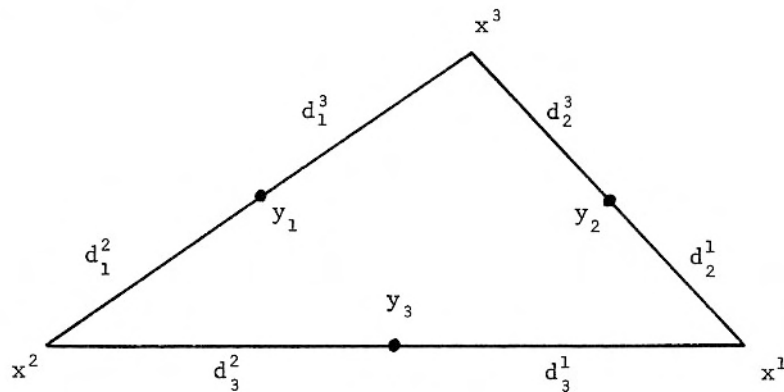


Figure 5